Rates and accumulated change

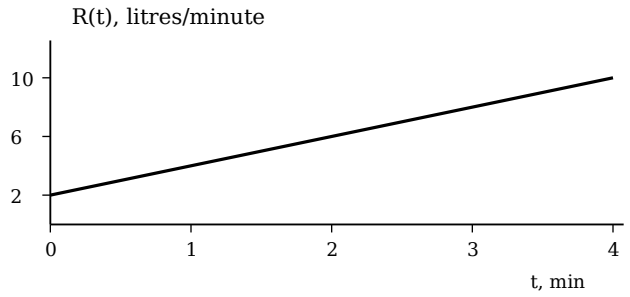
Original source page for reviewing narrow textbook prose.

---

14. A garden club is testing a small water pump. During the first four minutes, the flow rate rises steadily from two to ten litres per minute. Use the rule and graph below to estimate the water delivered in that interval. Divide the interval into four equal parts and use the left endpoint of each part. Then calculate the exact amount of water and explain why the estimate is smaller. State the units in your final answer, and include enough working for another student to check your method.

$$R(t) = 2 + 2t, \quad 0 \le t \le 4$$

$$\int_{0}^{4} (2 + 2t)\,dt \qquad \text{litres}$$



Original fixture. All words, equations, and diagrams were created for testing.